

Transverse micro-bunching of relativistic electron beams in structured solid targets

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The ability to generate dense, compact relativistic electron beams is of fundamental importance, with applications ranging from laboratory astrophysics to the study of beam-plasma instabilities. However, precisely controlling the transverse structure of such beams at extremely small spatial scales and at high densities of $\sim 10^{19-21} \text{ cm}^{-3}$ remains a major challenge. With the help of 2D Particle-In-Cell simulations, we demonstrate the controlled transverse micro-bunching of a 10 GeV electron beam down to submicron dimensions using structured solid targets. These targets consist of substrates patterned with arrays of hollow tubes—whose diameters span from the micrometer to nanometer range—aligned parallel to the beam propagation axis. In particular, we detail two regimes of interaction and explore the transition between both, while discussing the role of field and collisional ionization processes.

I. INTRODUCTION

Micrometer-scale relativistic electron beams, exhibiting extremely high charge densities on the order of $10^{19} - 10^{21} \text{ cm}^{-3}$, are central to a broad range of advanced plasma and laboratory astrophysics investigations. In laboratory astrophysics, such beams provide a controlled platform to initiate and to study **beam-plasma instabilities** [1–3], which remains an active topic of research [4–9], thereby offering insight into fundamental processes. These phenomena become especially important in solid-density plasmas [10], where interactions between dense relativistic beams and the background medium can drive the growth of intense electromagnetic fields. Furthermore, when relativistic beams interact with matter or strong electromagnetic waves, they can produce ultra-bright gamma-ray sources [11–13]. **High current electron beams are also envisioned to drive future soft x-ray light sources** [14, 15] and may even probe extreme conditions relevant for high-energy physics [16] and strong-field QED [17, 18]. However, currently available electron beams cannot reach the densities required to fully access these regimes [19].

In this context, we propose a method to further transversely compress existing relativistic beams by guiding them through hollow nanotubes. The main challenge lies in imprinting a transverse modulation on a GeV-class electron beam, with a modulation wavelength determined by the periodicity of the nanotube array, as illustrated in

Fig. 1. Accurately modeling this process using Particle-In-Cell (PIC) simulations is essential for understanding and ultimately optimizing the interaction.

Although the interaction of electron beams with hollow nanotube structures has received limited attention, the propagation of relativistic beams in hollow plasma channels (*i.e.*, single tubes) is an active area of research due to its relevance for positron acceleration [20, 21]. Existing studies [22–24] consistently show that the plasma screening of the beam’s electric field during propagation through a hollow nano-structure produces a strong transverse modulation. Consequently, accurate modeling of plasma ionization within the tubes is crucial, as it governs the field-screening dynamics and thus the amplitude of the beam-pinch effect.

In this paper, we start by illustrating in section II the transverse micro-bunching of a 10 GeV electron beam in hollow nano-tubes. Section III details the surface beam-plasma interaction that takes place for large tubes, and the volume beam-plasma interaction with thin tubes. We provide insight into the transition between the two regimes that takes place for hollow tubes with a diameter comparable to the plasma skin depth.

II. TRANSVERSE MICRO-BUNCHING

We ran 2D Particle-In-Cell simulations using the code Calder [25] and the initial setup is illustrated in Fig. 1(a).

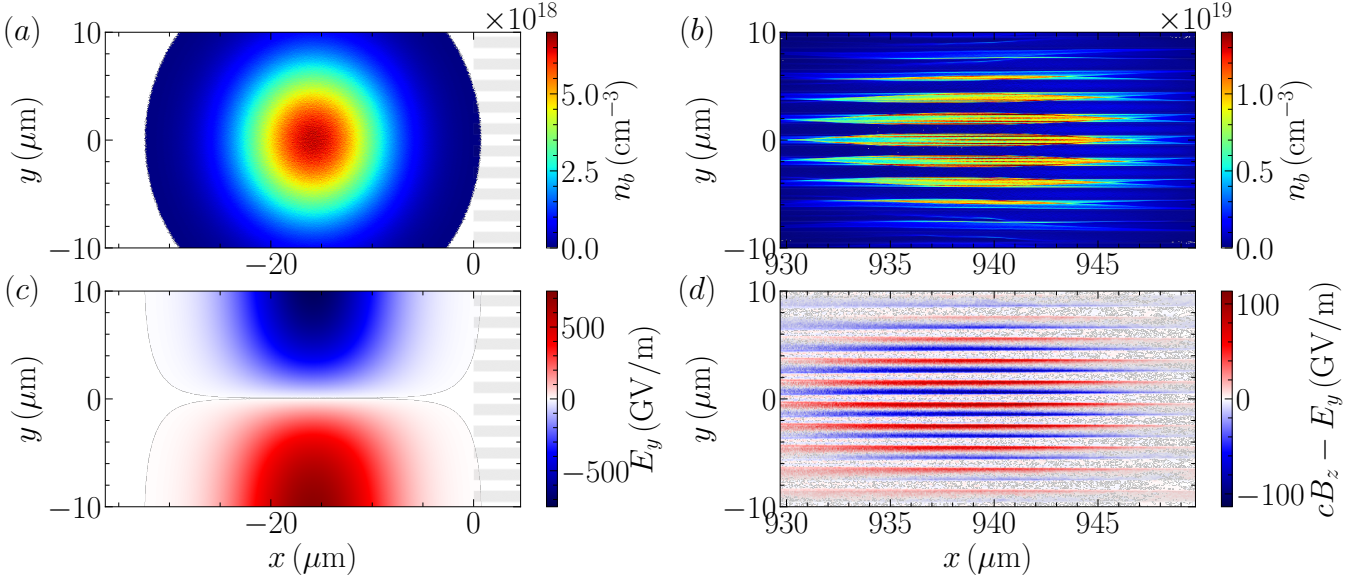


Figure 1. 2D particle-in-cell (PIC) simulation showing the transverse micro-bunching of a relativistic (10 GeV) electron beam within a hollow nanotube array. We show the electron beam density at (a) $t=0$ and (b) after $940 \mu\text{m}$ of propagation in the tubes. The plasma ions are represented in light gray color. (c) Transverse electric field E_y of the electron beam at $t=0$ (d) Transverse field experienced by electrons after $940 \mu\text{m}$ of propagation in the tubes.

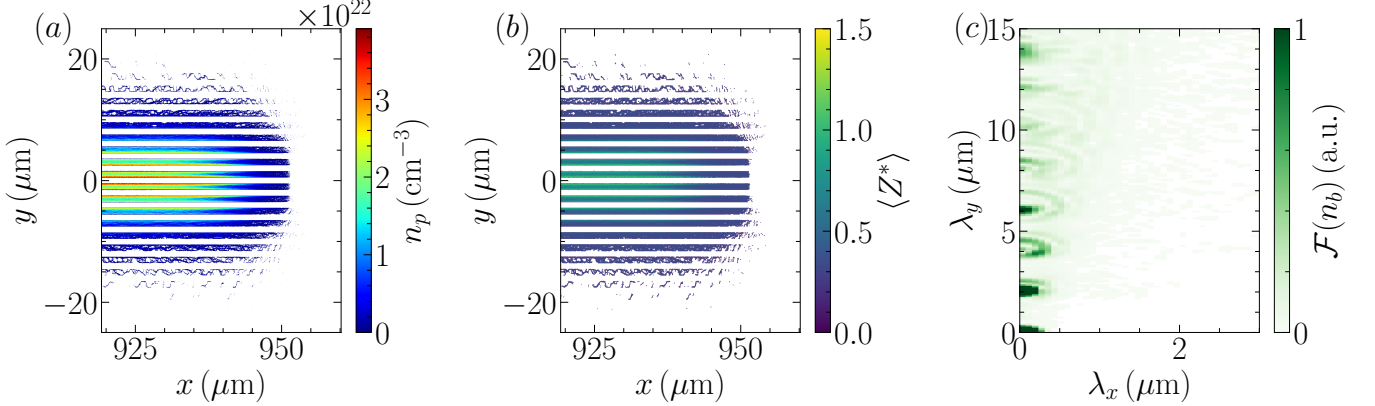


Figure 2. (a) Electron plasma density (b) average ionization degree per cell for Silicon ions (top) and Oxygen ions (bottom) (c) Fourier transform of the electron beam density showing the beam is transversely bunched at $2 \mu\text{m}$ wavelength, namely the period of the hollow nanotubes.

The beam propagates in direction x , and the transverse direction is denoted y . The density of electrons is $6 \times 10^{18} \text{ cm}^{-3}$, and it follows a Gaussian spatial profile with a standard deviation $\sigma = 5 \mu\text{m}$ in both x and y directions. It has a central energy of 10 GeV, and its transverse momentum distribution has a rms divergence of $15 \mu\text{rad}$. The plasma is made of neutral Silica, initialised with a number density of $1.6 \times 10^{22} \text{ cm}^{-3}$. The target is a solid material with cylindrical holes of nanometer to micrometer-diameter oriented parallel to the beam propagation axis. This nanostructure is deposited on a homogeneous substrate with a thickness of 1 mm. The hollow tubes have a thickness of 1 mm and a diameter denoted d ranging from $d = 20 \text{ nm}$ to $d = 6 \mu\text{m}$. We

introduce the periodicity of the tubes as the center-to-center distance between two consecutive hollow tubes. For all targets, the periodicity is twice the diameter of the hollow tubes, namely $2d$. In our simplified 2D geometry, this target therefore consists in layers of silica of thickness d , separated by layers of vacuum with size d . The domain has a size of $271 \times 314 \mu\text{m}^2$. For hollow nanotubes with a diameter above $d > 100 \text{ nm}$, we use a cell size of $10 \times 40 \text{ nm}^2$ and a time step of 32 as. For hollow nanotubes with a diameter of $d = 20 \text{ nm}$, we use a cell size of $40 \times 4 \text{ nm}^2$ and a time step of 9.4 as. The 2D simulations account both for field and collisional ionization. As the electron beam propagates at a relativistic velocity in the simulation domain, Numerical Cerenkov

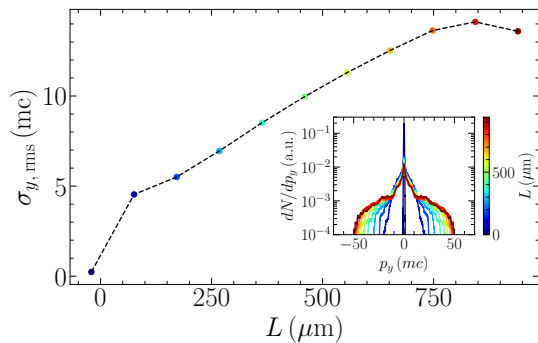


Figure 3. Standard deviation of the transverse momentum distribution of the electron beam as a function of its propagation distance in the hollow tubes. The inset displays the transverse momentum distribution at different propagation distances.

Radiation (NCR) [26] can be generated and strongly affect the beam evolution. To avoid it, we use a specific Maxwell solver designed to reduce NCR [27], as well as binomial filters to smooth the current and fields. We employ a 3rd order shape-factor for field interpolation and current gathering. For better energy conservation in the simulation domain, the scheme from Ref [28] for field interpolation is also used.

The electron beam initialized in the 2D simulations is represented in Fig. 1(a) before it enters the nanostructure. The self-fields of this beam are also initialized and illustrated in Fig. 1(c). In this figure, we consider the case of tubes with a diameter of $2\text{ }\mu\text{m}$, a periodicity of $2\text{ }\mu\text{m}$ and a thickness of $1\text{ }\mu\text{m}$. As the beam propagates in the hollow nanotubes, 2D PIC simulations reveal that it is bunched in the transverse direction. While the initial beam has a transverse size of $5\text{ }\mu\text{m}$, it is bunched into several co-propagating beams with a size of $\sim 1\text{ }\mu\text{m}$. It is worth noting that the peak density in the micro-bunches has increased by a factor of ~ 4 from $6 \times 10^{18}\text{ cm}^{-3}$ to $2.2 \times 10^{19}\text{ cm}^{-3}$ after $940\text{ }\mu\text{m}$ of propagation. The transverse bunching of the incident beam comes from the high-amplitude transverse fields growing in the plasma. To emphasize this effect, we plot the resulting transverse field $cB_z - E_y$ experienced by the electrons in Fig. 1(d). When the value is positive, electrons are deflected in the positive y direction; when negative, they are deflected in the negative y direction. The transverse pinching field reaches high amplitudes of $\sim 110\text{ GV/m}$, due to the screening of the electric field in the plasma. The details of the screening process are described in section III A and III B, depending on the regime of interaction.

Figure 2 provides more details about the interaction of the incident beam with the hollow nanotubes. As the beam propagates, its self-fields are sufficiently high to ionize the Silicon and Oxygen atoms in the Silica. We observe that the electron plasma density n_p represented in Fig. 2(a) is of the order of $10^{21} - 10^{22}\text{ cm}^{-3}$ and that it is larger at the back of the beam as compared to the

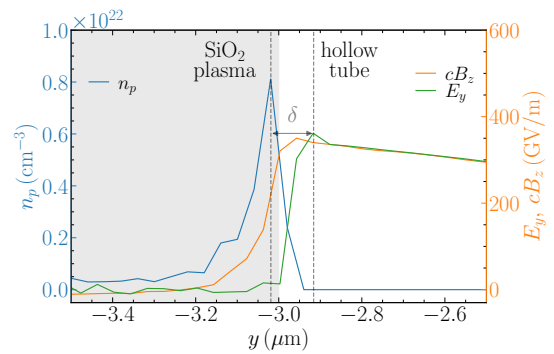


Figure 4. Line-out of the plasma electron density n_p , the transverse electric field E_y and magnetic field B_z in direction y , which is transverse to the beam propagation direction x . This figure is a zoom at the interface between the hollow tube and the SiO_2 plasma. The slight shift between the E_y and B_z fields in the plasma skin depth denoted δ induces a force pinching the incident electron beam in the transverse direction.

front. Indeed, a slice of the plasma located further from the beam's head experiences the propagation of a longer beam and may thus be more ionized than a slice of plasma at the head of the beam. The average ionization degree is shown in Fig. 2(b). We note that ions are ionized once at maximum. An average ionization degree lower than one means that only a fraction of atoms are ionized in each cell. The transverse bunching of the beam is analyzed with a Fourier transform in Fig 2(c). It reveals that the period of the bunching is $2\text{ }\mu\text{m}$, namely the periodicity of the hollow nanotube array. We also observe higher order modulations which are multiple of this period at $4, 6\text{ }\mu\text{m}$, etc.

Figure 3 illustrates the evolution of transverse momentum due to this transverse bunching. Each curve represents the distribution at a given propagation distance L in the nanostructure. Over the whole propagation, the standard deviation of the transverse momentum distribution increases from 0.3 mc to 18 mc , corresponding to an increase in rms divergence from $\sim 15\text{ }\mu\text{rad}$ to $\sim 0.9\text{ mrad}$.

III. DRIVING SURFACE OR VOLUME PLASMA PROCESSES WITH HOLLOW NANOTUBES

In this section, we report on the two regimes of interaction we identified during the propagation of a relativistic (10 GeV) electron beam with hollow nanotubes. In section III A, we analyse the surface interaction taking place when the diameter of the tube is larger than the plasma skin depth. In section III B, we analyse the volume interaction occurring as the diameter of the tube is smaller than the plasma skin depth.

A. Beam-surface interaction with large tubes

For this example, we consider the electron beam propagates from left to right along the x axis. We recall that its density is denoted n_b and its profile follows a Gaussian distribution with a standard deviation σ in all directions. In the [plasma](#), the electric field of the beam is screened on a typical scale-length equal to the plasma skin depth δ , while the magnetic field can extend on a larger length. The surface charge density growing in the [plasma](#) is $\delta \times n_p$ with n_p the plasma density in the tube. It compensates for the charge density of the electron beam [in the vacuum gaps](#), namely $\int_{-d/2}^{d/2} n_b(y) dy$. By equating these two quantities, we deduce the typical plasma skin depth we expect in the [plasma](#)

$$\delta = 2\sqrt{2}\sigma \frac{n_b}{n_p} \left[\operatorname{erf} \left(\frac{d}{2\sqrt{2}\sigma} \right) \right] \quad (1)$$

with $\operatorname{erf}(y) = 2/\sqrt{\pi} \int_0^y e^{-y^2} dy$ the error function. This estimate has a practical interest as it gives information on the regime of interaction between the beam and the [hollow nanostructure](#). In the limit $\delta \ll d$, the interaction takes place at the surface of the plasma. In the other limit $\delta \gg d$ the whole plasma volume experiences the fields of the beam.

We now test the estimate in Eq.(1) and show that it is coherent with the result of a 2D PIC simulation in Fig. 3. We consider the same electron beam as in the previous section with a density $n_b = 6 \times 10^{18} \text{ cm}^{-3}$ and a Gaussian density profile with $\sigma = 5 \mu\text{m}$ in all directions. It propagates along axis x and is centered at $y = 0$ in the transverse direction. The diameter of the tubes is chosen to be $d = 6 \mu\text{m}$, and the [first vacuum gap](#) initially extends between $y = \pm 3 \mu\text{m}$. Figure 4 is a line-out in the direction y transverse to the beam propagation axis. It illustrates the electric field E_y and magnetic field B_z as well as the electron density. The area with plasma ions is shaded in gray to highlight the position of the interface. The figure is a zoom extending close to one plasma-vacuum interface, for y in the range $-3.5 \mu\text{m} \leq y \leq -2.5 \mu\text{m}$. In the central part of the vacuum gap (for $y \geq -2.8 \mu\text{m}$), the electric and magnetic fields are the fields of the non-perturbed relativistic electron beam, so the resulting transverse Lorentz force is negligible. As a consequence, electrons in this region are not deflected. At the interface between the tube and vacuum for $y = -3 \mu\text{m}$, the plasma density in the skin depth rises to $8 \times 10^{21} \text{ cm}^{-3}$, to screen the electric field of the beam. The subsequent slight shift between the electric and magnetic field creates a positive Lorentz force (field amplitude is $\sim 100 \text{ s GV/m}$), large enough to push relativistic electrons from the beam in the positive y direction, namely from the plasma to the vacuum gap. Fig. 4 shows that the plasma density in the core of the tube (for $-3.5 \mu\text{m} \leq y \leq -3.2 \mu\text{m}$) is $n_p = 5 \times 10^{20} \text{ cm}^{-3}$. Using this value, the estimate of the plasma skin depth from Eq. (1) is $\sim 77 \text{ nm}$, which is coherent with the value from the 2D

PIC simulation ($\sim 100 \text{ nm}$).

B. Volume beam-plasma interaction with thin tubes

A volume plasma response can be induced for thin hollow tubes as their diameter becomes smaller than the plasma skin depth. In this case, the number of electrons in the plasma is insufficient to completely screen the field of the incident electron beam. As a result, electrons are expelled from the plasma and fill the whole volume of the nanostructure.

Figure 5 details the interaction of the incident beam with the plasma for a 2D PIC simulation accounting only for field ionization and ignoring collisional ionization. The hollow tubes have a diameter of 20 nm and a periodicity of 40 nm . The figure is a zoom on the central part of the beam, and the whole simulation domain is larger as stated in section II. For such a small diameter, there are approximately one hundred hollow tubes per plasma skin depth. We compute the average plasma density and degree of ionization from the data in Figs 5(b) and 5(d). Although we are averaging quantities that are not locally homogeneous, the outcome values are relevant to discuss whether plasma waves can be sustained. The average plasma density is $n_p = 2.5 \times 10^{20} \text{ cm}^{-3}$, corresponding to an average degree of ionization $Z^* = 0.5$, and a wavelength of $\lambda_p = 2.1 \mu\text{m}$. We recall that the electron beam driver has a density of $6 \times 10^{18} \text{ cm}^{-3}$ and a length of $5 \mu\text{m}$. In addition, Fig. 5(b) shows that the plasma density gradient is very sharp. The conditions to drive a linear wake-field are therefore met, as confirmed in Fig. 5(c). In the latter, we represent the longitudinal electric field in the plasma with the red blue color bar. The measured wavelength of the longitudinal electric field is $2.1 \mu\text{m}$, which is coherent with the average plasma density $n_p = 2.5 \times 10^{20} \text{ cm}^{-3}$. We note that the electron beam density (see Fig. 5(a)) is modulated longitudinally at the plasma wavelength, and also transversely at the period of the nanotubes (40 nm).

However, the formation of a plasma wave is completely damped when 2D simulations account for collisional ionization. We ran the same simulation as in Fig. 5 for tubes with a diameter of 20 nm and we added the process of collisional ionization. The results are reported in Fig. 6. When we account for field and collisional ionization, the average plasma density increases by a factor of ~ 20 , from $2.5 \times 10^{20} \text{ cm}^{-3}$ to $5.7 \times 10^{21} \text{ cm}^{-3}$, as computed from the data in Fig. 6(b). This is corroborated by the increase of the average degree of ionization from 0.5 to 1.3 (Fig. 6(d)). From this higher ionization degree, we can infer that the secondary electrons emitted from field ionization collide with the atoms and ionize them to a higher charge state. The plasma wavelength is now $0.4 \mu\text{m}$, and the length of the driver ($5 \mu\text{m}$) is too large to be resonant and excite a plasma wave. With collisional ionization, we also observe in 6(b) that the plasma density

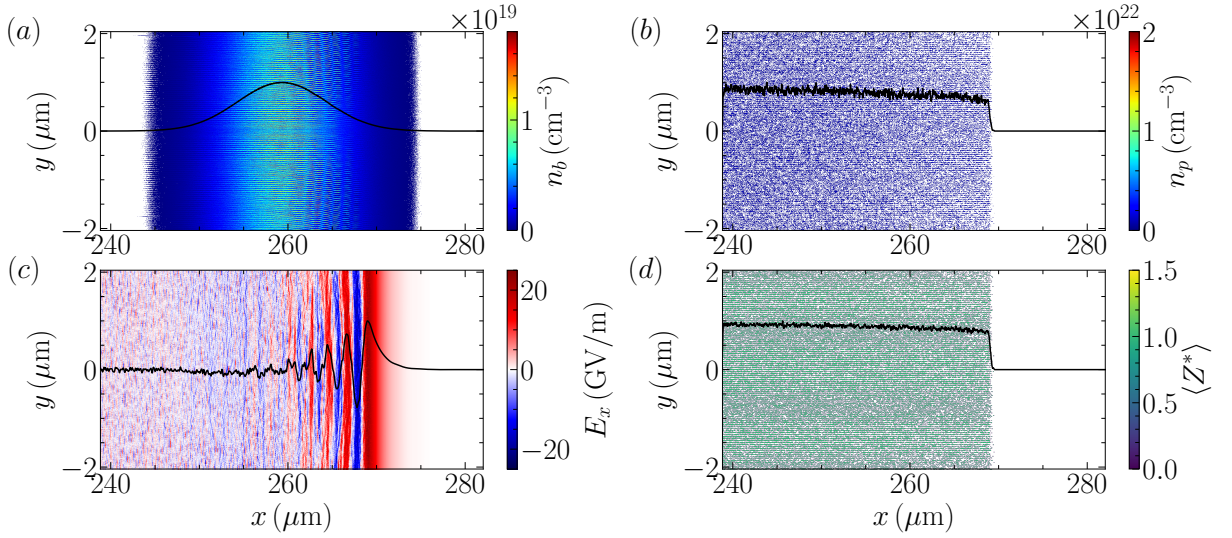


Figure 5. 2D PIC simulations with field ionization only. (a) Incident electron beam density (b) background electron density (c) longitudinal electric field (d) average ionization degree in each cell. For each figure, the black curve is an average of the quantity along the transverse direction y . This curve is normalized to 1.

gradient is more gradual than in Fig. 5(b) without collisional ionization. As a consequence of the non-matching resonance condition and the longer plasma density gradient, no plasma wave is excited. This is confirmed in Fig. 6(c) where the longitudinal electric field only shows a single peak from a local charge separation, but does not have a characteristic modulation of a plasma wave. As a consequence, the electron beam does not develop a longitudinal density modulation as it did without collisional ionization. It is still modulated transversely at the period of the nano-structure (40 nm).

C. Transition from volume to surface interaction

Now that we exemplified the two regimes of interaction during the propagation of an electron beam in a nano-tube array, we provide more insight on the transition from one to the other.

Depending on the regime of interaction, the final spread of the incident electron beam shows different features. Figure 7 details the transverse momentum distribution of the beam as a function of the propagation distance in the nano-tubes. For nanotubes with a diameter of ≤ 200 nm, the distribution is Gaussian with a standard deviation of $\sim 5mc$. We note that this distribution is established early in the interaction and barely changes during the whole propagation of the beam. The reason is that in this regime of volume beam-plasma interaction the plasma is quickly homogenized, as exemplified in Fig. 6(b). As there is no plasma wave formation and no instabilities, the spread of the beam remains constant. On the contrary, we observe a different behavior when tubes are large enough to set the interaction in the surface beam-plasma regime, i.e. for the tubes with a

larger diameter $d \geq 500$ nm. The transverse spread of the beams grows linearly with the propagation distance. We showed qualitatively in Fig. 4 that this transverse bunching comes from the strong transverse force induced by the screening of the electric field in the plasma. The rate of increase is higher as d increases from 0.5 to $2 \mu\text{m}$. For the largest tube diameter $d = 6 \mu\text{m}$, comparable to the incident beam transverse size, this rate suddenly drops. A quantitative explanation of this tendency would require a theoretical estimation of the transverse force as a function of the diameter of the tubes. This remains challenging as it requires to obtain an estimate for the plasma density and ionization in the Silica plasma.

Figure 8 summarizes how the rms value of the transverse momentum of the beam evolves as a function of the diameter of the nanotubes. We performed 2D simulations with different ionization processes and conclude that collisional ionization plays a significant role on the final momentum spread of the electron beam. The transition between the two regimes of interaction (either in surface or in volume) takes place for a tube diameter $200 \leq d \leq 500$ nm. As in Fig. 7, we observe that the transverse spread of the beam does not vary significantly in the regime of volume beam-plasma interaction ($d \leq 200$ nm). We recall the reason is that in all these cases, the beam-plasma parameters are such that no instability nor collective plasma process are triggered, and we chose the nano-tube arrays such that they have the same average density. For nano-tubes with a larger diameter ($d \geq 500$ nm), the beam electric field is completely screened, leading to a deflection of electrons in the direction transverse to their propagation. This leads to a larger momentum spread for the electrons (Fig. 8).

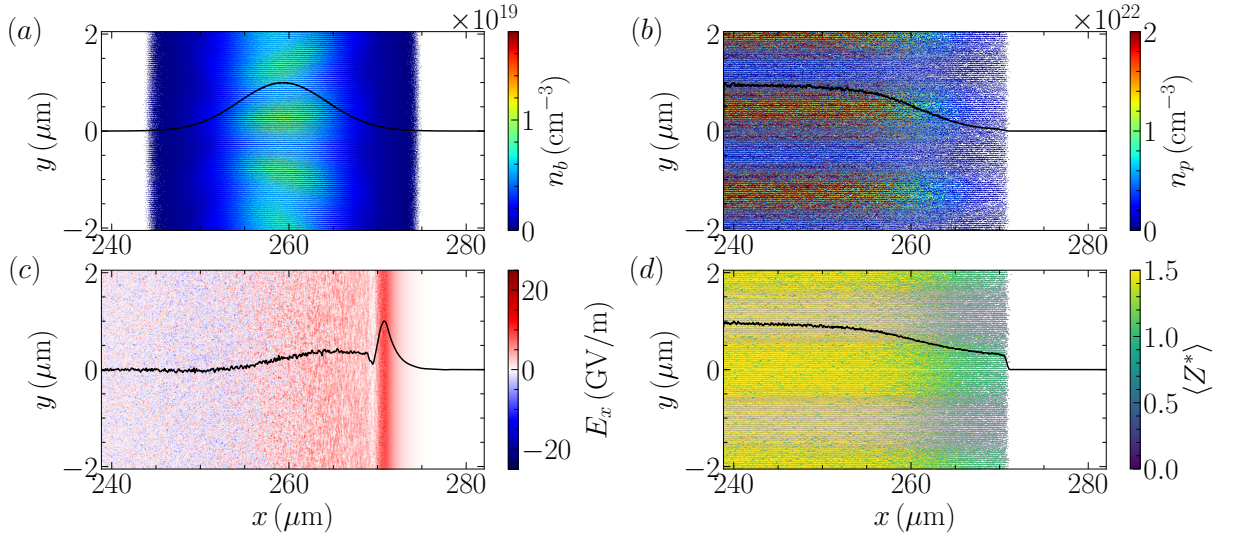


Figure 6. 2D PIC simulations with field and collisional ionization. (a) Incident electron beam density (b) background electron density (c) longitudinal electric field (d) average ionization degree in each cell. For each figure, the black curve is an average of the quantity along the transverse direction y . This curve is normalized to 1.

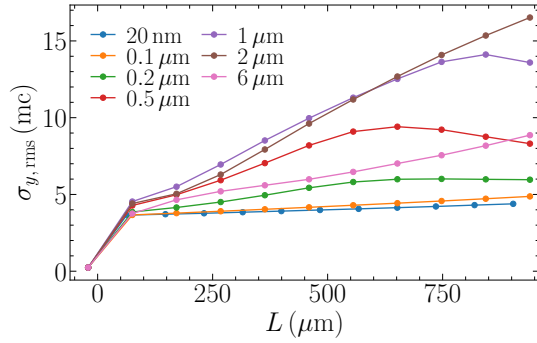


Figure 7. Standard deviation of the transverse momentum distribution of a relativistic electron beam interacting with a structured hollow nano-tube target.

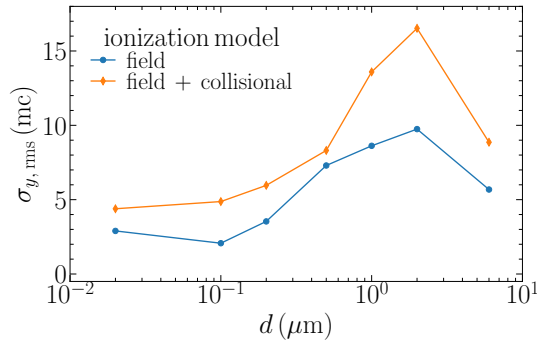


Figure 8. Standard deviation of the transverse momentum distribution of the electron beam as a function of the tube's diameter d after $940\mu\text{m}$ of propagation in the tubes. 2D simulations are performed with field ionization only (circles) and also with collisional ionization (diamonds).

IV. CONCLUSION

Using 2D Particle-In-Cell simulations, we examined the transverse micro-bunching of a relativistic electron beam propagating through a nanostructured target. Within the material, atoms are ionized either by the strong electromagnetic field of the incident beam or through collisions with thermal plasma electrons. This ionization leads to a gradual screening of the beam's electric field in the plasma, thereby inducing transverse bunching of the relativistic electrons. Because the screening evolves over a characteristic length set by the plasma skin depth, the interaction is found to operate in two distinct regimes: a surface-dominated regime for tube diameters exceeding the skin depth, and a volume-dominated regime for smaller tubes. We demonstrated that for thin tubes, the rapidly homogenized plasma may behave as a uniform plasma and sustain the propagation of plasma waves. However, it remains crucial to account for collisional ionization as we proved it may damp the growth of plasma waves.

As the electron beam undergoes transverse bunching within the nanotube array, each resulting micro-bunch attains a higher density and reduced transverse size compared with the initial beam. Although we quantified this effect using 2D PIC simulations, a full theoretical characterization of the underlying dynamics remains an important direction for future work.

V. ACKNOWLEDGMENTS

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- [1] J. Warwick, T. Dzelzainis, M. E. Dieckmann, W. Schumaker, D. Doria, L. Romagnani, K. Poder, J. M. Cole, A. Alejo, M. Yeung, K. Krushelnick, S. P. D. Mangles, Z. Najmudin, B. Reville, G. M. Samarin, D. D. Symes, A. G. R. Thomas, M. Borghesi, and G. Sarri, *Phys. Rev. Lett.* **119**, 185002 (2017).
 - [2] C. D. Arrowsmith, P. Simon, P. J. Bilbao, A. F. A. Bott, S. Burger, H. Chen, F. D. Cruz, T. Davenne, I. Efthymiopoulos, D. H. Froula, A. Goillot, J. T. Gudmundsson, D. Haberberger, J. W. D. Halliday, T. Hodge, B. T. Huffman, S. Iaquina, F. Miniati, B. Reville, S. Sarkar, A. A. Schekochihin, L. O. Silva, R. Simpson, V. Stergiou, R. M. G. M. Trines, T. Vieu, N. Charitonidis, R. Bingham, and G. Gregori, *Nature Communications* **15**, 5029 (2024).
 - [3] C. D. Arrowsmith, F. Miniati, P. J. Bilbao, P. Simon, A. F. A. Bott, S. Burger, H. Chen, F. D. Cruz, T. Davenne, A. Dyson, I. Efthymiopoulos, D. H. Froula, A. Goillot, J. T. Gudmundsson, D. Haberberger, J. W. D. Halliday, T. Hodge, B. T. Huffman, S. Iaquina, G. Marshall, B. Reville, S. Sarkar, A. A. Schekochihin, L. O. Silva, R. Simpson, V. Stergiou, R. M. G. M. Trines, T. Vieu, N. Charitonidis, R. Bingham, and G. Gregori, *Proceedings of the National Academy of Sciences* **122**, e2513365122 (2025).
 - [4] R. N. Sudan, in *Handbook of Plasma Physics*, Vol. 2, edited by M. N. Rosenbluth and R. Z. Sagdeev (North-Holland Physics Publishing, Amsterdam, 1984) Chap. 6.3, pp. 337–382.
 - [5] L. O. Silva, R. A. Fonseca, J. W. Tonge, W. B. Mori, and J. M. Dawson, *Physics of Plasmas* **9**, 2458 (2002).
 - [6] A. Bret, M.-C. Firpo, and C. Deutsch, *Phys. Rev. Lett.* **94**, 115002 (2005).
 - [7] A. Bret, L. Gremillet, and D. Bénisti, *Phys. Rev. E* **81**, 036402 (2010).
 - [8] B. Allen, V. Yakimenko, M. Babzien, M. Fedurin, K. Kusche, and P. Muggli, *Phys. Rev. Lett.* **109**, 185007 (2012).
 - [9] P. San Miguel Claveria, X. Davoine, J. R. Peterson, M. Gilljohann, I. Andriyash, R. Ariniello, C. Clarke, H. Ekerfelt, C. Emma, J. Faure, S. Gessner, M. J. Hogan, C. Joshi, C. H. Keitel, A. Knetsch, O. Kononenko, M. Litos, Y. Mankovska, K. Marsh, A. Matheron, Z. Nie, B. O'Shea, D. Storey, N. Vafaei-Najafabadi, Y. Wu, X. Xu, J. Yan, C. Zhang, M. Tamburini, F. Fiuza, L. Gremillet, and S. Corde, *Phys. Rev. Res.* **4**, 023085 (2022).
 - [10] A. Benedetti, M. Tamburini, and C. H. Keitel, *Nature Photonics* **12**, 319 (2018).
 - [11] A. Sampath, X. Davoine, S. Corde, L. Gremillet, M. Gilljohann, M. Sangal, C. H. Keitel, R. Ariniello, J. Cary, H. Ekerfelt, C. Emma, F. Fiuza, H. Fujii, M. Hogan, C. Joshi, A. Knetsch, O. Kononenko, V. Lee, M. Litos, K. Marsh, Z. Nie, B. O'Shea, J. R. Peterson, P. S. M. Claveria, D. Storey, Y. Wu, X. Xu, C. Zhang, and M. Tamburini, *Phys. Rev. Lett.* **126**, 064801 (2021).
 - [12] A. S. Samsonov and I. Y. Kostyukov, *Optics and Spectroscopy* **130**, 219 (2022).
 - [13] L.-J. Cui, K.-J. Wei, C. Lv, F. Wan, Y. I. Salamin, L.-F. Cao, and J.-X. Li, *Phys. Rev. A* **111**, 023505 (2025).
 - [14] X. Xu, D. B. Cesar, S. Corde, V. Yakimenko, M. J. Hogan, C. Joshi, A. Marinelli, and W. B. Mori, *Phys. Rev. Lett.* **126**, 094801 (2021).
 - [15] C. Emma, X. Xu, A. Fisher, R. Robles, J. P. MacArthur, J. Cryan, M. J. Hogan, P. Musumeci, G. White, and A. Marinelli, *APL Photonics* **6**, 076107 (2021).
 - [16] V. Yakimenko, S. Meuren, F. Del Gaudio, C. Baumann, A. Fedotov, F. Fiuza, T. Grismayer, M. J. Hogan, A. Pukhov, L. O. Silva, and G. White, *Phys. Rev. Lett.* **122**, 190404 (2019).
 - [17] A. Gonoskov, T. G. Blackburn, M. Marklund, and S. S. Bulanov, *Rev. Mod. Phys.* **94**, 045001 (2022).
 - [18] A. Matheron, P. San Miguel Claveria, R. Ariniello, H. Ekerfelt, F. Fiuza, S. Gessner, M. F. Gilljohann, M. J. Hogan, C. H. Keitel, A. Knetsch, M. Litos, Y. Mankovska, S. Montefiori, Z. Nie, B. O'Shea, J. R. Peterson, D. Storey, Y. Wu, X. Xu, V. Zakharova, X. Davoine, L. Gremillet, M. Tamburini, and S. Corde, *Communications Physics* **6**, 141 (2023).
 - [19] C. Emma, N. Majernik, K. K. Swanson, R. Ariniello, S. Gessner, R. Hessami, M. J. Hogan, A. Knetsch, K. A. Larsen, A. Marinelli, B. O'Shea, S. Perez, I. Rajkovic, R. Robles, D. Storey, and G. Yocky, *Phys. Rev. Lett.* **134**, 085001 (2025).
 - [20] S. Gessner, E. Adli, J. M. Allen, W. An, C. I. Clarke, C. E. Clayton, S. Corde, J. P. Delahaye, J. Frederico, S. Z. Green, C. Hast, M. J. Hogan, C. Joshi, C. A. Lindström, N. Lipkowitz, M. Litos, W. Lu, K. A. Marsh, W. B. Mori, B. O'Shea, N. Vafaei-Najafabadi, D. Walz, V. Yakimenko, and G. Yocky, *Nature Communications* **7**, 11785 (2016).
 - [21] T. Silva, L. D. Amorim, M. C. Downer, M. J. Hogan, V. Yakimenko, R. Zgadzaj, and J. Vieira, *Phys. Rev. Lett.* **127**, 104801 (2021).
 - [22] X.-L. Zhu, W.-Y. Liu, M. Chen, S.-M. Weng, D. Wu, T.-P. Yu, W.-M. Wang, Z.-M. Sheng, and J. Zhang, *New Journal of Physics* **25**, 093016 (2023).

- [23] X.-L. Zhu, W.-Y. Liu, T.-P. Yu, M. Chen, S.-M. Weng, W.-M. Wang, and Z.-M. Sheng, *Phys. Rev. Lett.* **132**, 235001 (2024).
- [24] A. Bonatto, G. Xia, O. Apsimon, C. Bontoiu, E. Kukstas, V. Rodin, M. Yadav, C. P. Welsch, and J. Resta-López, *Physics of Plasmas* **30**, 033105 (2023).
- [25] E. Lefebvre, N. Cochet, S. Fritzler, V. Malka, M.-M. Aléonard, J.-F. Chemin, S. Darbon, L. Disdier, J. Faure, A. Fedotoff, O. Landoas, G. Malka, V. Méot, P. Morel, M. R. L. Gloahec, A. Rouyer, C. Rubbe-lynck, V. Tikhonchuk, R. Wrobel, P. Audebert, and C. Rousseaux, *Nuclear Fusion* **43**, 629 (2003).
- [26] B. B. Godfrey, *Journal of Computational Physics* **15**, 504 (1974).
- [27] R. Lehe, A. Lifschitz, C. Thaury, V. Malka, and X. Davoine, *Phys. Rev. ST Accel. Beams* **16**, 021301 (2013).
- [28] I. V. Sokolov, *Computer Physics Communications* **184**, 320–328 (2013).